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System and method using internal cancellation magnets to control magnetic wheel adhesion

Abstract

A system and method control magnetic adhesion of a wheel to a surface using internal cancellation magnets. An inner annular disc of the wheel comprises a non-magnetic material and has first and second apertures which retain magnets while an outer annular disc comprises a ferromagnetic material and is disposed on a side of the inner annular disc and has a non-magnetic isolator ring which extends in a serpentine manner. In one configuration, the curves of the serpentine isolator ring isolate the first magnets from the second magnets. In another configuration, the outer annular disc rotates relative to the inner annular disc to dispose the curves of the serpentine isolator ring in a second position to allow magnetic interaction between the first and second magnets to generate a second magnetic flux between the second magnets and the ferromagnetic surface to thereby decrease the adhesion of the wheel to the ferromagnetic surface.

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Background/Summary

FIELD OF THE DISCLOSURE

(1) The present disclosure relates generally to magnetized wheels, and, more particularly, to a system and method using internal cancellation magnets to control magnetic adhesion of a wheel to a ferromagnetic surface.

BACKGROUND OF THE DISCLOSURE

(2) Magnetic wheels enable vehicles to climb and drive on ferromagnetic structures. For example, an unmanned aerial vehicle (UAV) can fly to a point on a ferromagnetic structure, perch at that point, and utilize magnetic wheels to adhere to the ferromagnetic structure. The magnetic adhesion is the result of magnetic flux passing through the surface from the magnet north pole to the magnetic south pole of a magnet in the wheel. Having a strong magnetic grip to the ferromagnetic surface is essential to prevent the vehicle from disengaging inadvertently and from falling from the ferromagnetic surface. However, a strong pulling force is required to overcome the magnetic adhesion to disengage the vehicle from the ferromagnetic surface. In order to enable vehicles to obtain a strong magnetic grip as well as easy disengagement, incorporation of a magnetic switch into the wheel is desirable.

SUMMARY OF THE DISCLOSURE

(3) According to an embodiment consistent with the present disclosure, a system and method using internal cancellation magnets to control magnetic adhesion of a wheel to a ferromagnetic surface.

(4) In an embodiment, a wheel configured to adhere magnetically to a ferromagnetic surface, comprising an inner annular disc and a pair of outer annular discs. The inner annular disc is composed of a non-magnetic material and has a first outer circumferential periphery, a first central axial bore, a first plurality of apertures, and a second plurality of apertures. The first central axial bore is configured to retain an axle. The first plurality of apertures is disposed adjacent to the first central axial bore and configured to retain a first plurality of magnets. The second plurality of apertures disposed adjacent to the outer circumferential periphery and configured to retain a second plurality of magnets.

(5) The pair of outer annular discs are composed of a ferromagnetic material and are disposed on either side of the inner annular disc. Each outer annular disc has a second outer circumferential

periphery, a second central axial bore, an inner circumferential periphery, and an isolator ring. The second central axial bore is configured to retain an axle. The inner circumferential periphery is disposed adjacent to the second central axial bore. The isolator ring is composed of a non-magnetic material and defines a plurality of curves extending in a serpentine manner circumferentially around the inner circumferential periphery intermediate of the second outer circumferential periphery and the second central axial bore, with the curves of the serpentine isolator ring disposed between the inner circumferential periphery and the second outer circumferential periphery. In a first configuration, the curves of the serpentine isolator ring are disposed in a first position relative to the second plurality of magnets to isolate the first plurality of magnets from the second plurality of magnets, thereby generating a first magnetic flux between the second plurality of magnets and the ferromagnetic surface to increase the adhesion of the wheel to the ferromagnetic surface. In a second configuration, at least one outer annular disc is rotated about the axle relative to the inner annular disc to dispose the curves of the serpentine isolator ring in a second position relative to the second plurality of magnets to allow magnetic interaction between the first plurality of magnets and the second plurality of magnets, thereby generating a second magnetic flux between the second plurality of magnets and the ferromagnetic surface to decrease the adhesion of the wheel to the ferromagnetic surface. The second magnetic flux is less than the first magnetic flux.

(6) Each of the first plurality of magnets has a first polarity, and each of the second plurality of magnets has a second polarity opposite to the first polarity. In the second configuration, the first plurality of magnets at least partially cancels the magnetic flux of the second plurality of magnets. The first plurality of magnets are permanent magnets. The second plurality of magnets are permanent magnets. The first and second pluralities of apertures are cylindrical. The first and second pluralities of magnets are cylindrical.

(7) In another embodiment, a wheel is configured to adhere magnetically to a ferromagnetic surface, and comprises a first annular disc and a second annular disc. The first annular disc is composed of a non-magnetic material and has a first outer circumferential periphery, a first central axial bore, a first plurality of apertures, and a second plurality of apertures. The first central axial bore is configured to retain an axle. The first plurality of apertures are disposed adjacent to the first central axial bore and are configured to retain a first plurality of magnets. The second plurality of apertures are disposed adjacent to the outer circumferential periphery and are configured to retain a second plurality of magnets.

(8) The second annular disc is composed of a ferromagnetic material and is disposed on one side of the first annular disc, with the outer annular disc having a second outer circumferential periphery, a second central axial bore, an inner circumferential periphery, and an isolator ring. The second central axial bore is configured to retain an axle. The inner circumferential periphery is disposed adjacent to the second central axial bore. The isolator ring is composed of a non-magnetic material and defines a plurality of curves extending in a serpentine manner circumferentially around the inner circumferential periphery intermediate of the second outer circumferential periphery and the second central axial bore, with the curves of the serpentine isolator ring disposed between the inner circumferential periphery and the second outer circumferential periphery. In a first configuration, the curves of the serpentine isolator ring are disposed in a first position relative to the second plurality of magnets to isolate the first plurality of magnets from the second plurality of magnets, thereby generating a first magnetic flux between the second plurality of magnets and the ferromagnetic surface to increase the adhesion of the wheel to the ferromagnetic surface. In a second configuration, the second annular disc is rotated about the axle relative to the first annular disc to dispose the curves of the serpentine isolator ring in a second position relative to the second plurality of magnets to allow magnetic interaction between the first plurality of magnets and the second plurality of magnets, thereby generating a second magnetic flux between the second plurality of magnets and the ferromagnetic surface to decrease the adhesion of the wheel to the ferromagnetic surface. The second magnetic flux is less than the first magnetic flux.

(9) Each of the first plurality of magnets has a first polarity, and each of the second plurality of magnets has a second polarity opposite to the first polarity. In the second configuration, the first plurality of magnets at least partially cancels the magnetic flux of the second plurality of magnets. The first plurality of magnets are permanent magnets. The second plurality of magnets are permanent magnets. The first and second pluralities of apertures are cylindrical. The first and second pluralities of magnets are cylindrical.

(10) In a further embodiment, a method of adhering a wheel magnetically to a ferromagnetic surface comprises providing a wheel having a first annular disc and a second annular disc, wherein the first annular disc is composed of a non-magnetic material and has a first outer circumferential periphery, a first central axial bore, a first plurality of apertures, and a second plurality of apertures. The first central axial bore is configured to retain an axle. The first plurality of apertures are disposed adjacent to the first central axial bore and are configured to retain a first plurality of magnets. A second plurality of apertures is disposed adjacent to the outer circumferential periphery and is configured to retain a second plurality of magnets. The second annular disc is composed of a ferromagnetic material and is disposed on one side of the first annular disc, with the outer annular disc having a second outer circumferential periphery, a second central axial bore, an inner circumferential periphery, and an isolator ring. The second central axial bore is configured to retain an axle. The inner circumferential periphery is disposed adjacent to the second central axial bore. The isolator ring is composed of a non-magnetic material and defines a plurality of curves extending in a serpentine manner circumferentially around the inner circumferential periphery intermediate of the second outer circumferential periphery and the second central axial bore, with the curves of the serpentine isolator ring disposed between the inner circumferential periphery and the second outer circumferential periphery.

(11) The method also comprises disposing the first and second annular discs in a first configuration wherein the curves of the serpentine isolator ring are disposed in a first position relative to the second plurality of magnets, isolating the first plurality of magnets magnetically from the second plurality of magnets, and generating a first magnetic flux between the second plurality of magnets and the ferromagnetic surface thereby increasing the adhesion of the wheel to the ferromagnetic surface.

(12) The method further comprises disposing the first and second annular discs in a second configuration, wherein the second annular disc is rotated about the axle relative to the first annular disc, disposing the curves of the serpentine isolator ring in a second position relative to the second plurality of magnets, allowing magnetic interaction between the first plurality of magnets and the second plurality of magnets, and generating a second magnetic flux between the second plurality of magnets and the ferromagnetic surface to decrease the adhesion of the wheel to the ferromagnetic surface, wherein the second magnetic flux is less than the first magnetic flux.

(13) Each of the first plurality of magnets has a first polarity, and each of the second plurality of magnets has a second polarity opposite to the first polarity. The first plurality of magnets are permanent magnets. The second plurality of magnets are permanent magnets. The first and second pluralities of apertures are cylindrical.

(14) Any combinations of the various embodiments and implementations disclosed herein can be used in a further embodiment, consistent with the disclosure. These and other aspects and features can be appreciated from the following description of certain embodiments presented herein in accordance with the disclosure and the accompanying drawings and claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a diagram of a wheel, according to an embodiment.

- (2) FIG. 2 is a diagram of the wheel of FIG. 1 with parts separated.
- (3) FIG. 3 illustrates outer discs of the wheel of FIG. 1 rotating in a common direction relative to an inner disc.
- (4) FIG. 4 illustrates the outer discs of the wheel of FIG. 1 rotating in opposite directions relative to the inner disc.
- (5) FIG. 5 illustrates the inner disc of the wheel of FIG. 1 rotating relative to both of the outer discs.
- (6) FIG. 6 is a cross-sectional view of the wheel of FIG. 1 in a first configuration.
- (7) FIG. 7 illustrates the magnetic flux of the wheel in the first configuration of FIG. 6 along lines 7-7.
- (8) FIG. 8 is a cross-sectional view of the wheel of FIG. 1 in a second configuration.
- (9) FIG. 9 illustrates the magnetic flux of the wheel in the second configuration of FIG. 8 along lines 9-9.
- (10) FIG. 10 is a flowchart of the method of operation of the wheel of FIG. 1.
- (11) It is noted that the drawings are illustrative and are not necessarily to scale.

DETAILED DESCRIPTION OF CERTAIN EMBODIMENTS OF THE DISCLOSURE

- (12) Example embodiments consistent with the teachings included in the present disclosure are directed to a system and method using internal cancellation magnets to control magnetic adhesion of a wheel to a ferromagnetic surface.
- (13) Referring to FIGS. 1-9, the wheel 10 is configured to roll on a surface 12. Using the system and method described below, when the surface 12 is ferromagnetic, a magnetic flux generated by the wheel 10 can be controlled to increase or decrease magnetic adhesion of the wheel 10 to the ferromagnetic surface 12. As shown in FIGS. 1-2, the wheel 10 has an inner disc 14 having a pair of planar sides 16. The inner disc 14 is composed of a non-magnetic material. For example, the inner disc 14 can be composed of plastic. The inner disc 14 is generally annular with a central axial bore 18 partially surrounded by a disc retainer 20, 22 extending perpendicularly from at least one of the planar sides 16. The central axial bore 18 is configured to receive an axle to roll the wheel 10 about the axle on the surface 12. A plurality of apertures 24, 26 extend perpendicularly and at least partially into at least one of the planar sides 16. A first plurality of apertures 24 are disposed adjacent to the first central axial bore and configured to retain a first plurality of magnets 28 as inner magnets. A second plurality of apertures 26 are disposed adjacent to the outer circumferential periphery and configured to retain a second plurality of magnets 30 as outer magnets. The magnets 28, 30 are sized and dimensioned to be retained in the respective apertures 24, 26. In an example embodiment, the apertures 24, 26 are cylindrical, and the magnets 28, 30 are also cylindrical.
- (14) The wheel 10 also has at least one outer disc 32, 34 disposed adjacent to a respective planar side of the inner disc 14. Each outer disc 32, 34 is composed of a ferromagnetic material. For example, each outer disc 32, 34 can be composed of steel. Alternatively, each outer disc 32, 34 can be composed of nickel. In another alternative embodiment, each outer disc 32, 34 can be composed of cobalt. The outer disc 32, 34 can also be composed of other ferromagnetic materials. Each outer disc 32, 34 has a central axial bore 36 configured to receive a respective disc retainer 20, 22 through which an axle passes. The central axial bore 36 of each outer disc 32, 34 has a rotation stopper 38 for engaging radial sides 40, 42 of a respective disc retainer 20, 22. The radial sides 40, 42 limit the rotation of each outer disc 32, 34 relative to the inner disc 14 to a predetermined angle. The predetermined angle is equal to $180^\circ/(\text{the number of outer magnets on an outer circumferential periphery of the inner disc})$. For example, in an embodiment with eight outer magnets, the predetermined angle can be about $180^\circ/8$, which is about 22.5° . Using a different number of outer magnets would change the predetermined angle. For example, for ten outer magnets, the rotation angle can be about $180^\circ/10$, which is about 18° . It is also understood that other sizes and dimensions of the outer diameters of the discs as well as the aperture sizes are contemplated.
- (15) Referring again to FIGS. 1-2, each outer disc 32, 34 has an outer circumferential periphery 44

and an inner circumferential periphery **46** disposed adjacent to the central axial bore **36**. An isolator ring **48** composed of a non-magnetic material and defining a plurality of curves extends in a serpentine manner circumferentially around the inner circumferential periphery **46** intermediate of the outer circumferential periphery **44** and the central axial bore **36**, with the curves of the serpentine isolator ring **48** disposed between the inner circumferential periphery **46** and the outer circumferential periphery **44**. The isolator ring **48** can be composed of plastic.

(16) As described above, rotation of each outer disc **32**, **34** can be performed relative to the inner disc **14** to within a predetermined angle. FIG. 3 illustrates outer discs **32**, **34** of the wheel **10** rotating in a common direction relative to the inner disc **14**. As shown in FIG. 3, the inner disc **14** can be motionless as the outer discs **32**, **34** rotate in a common direction. FIG. 4 illustrates the outer discs **32**, **34** of the wheel **10** rotating in opposite directions relative to the inner disc **14**. As shown in FIG. 4, the inner disc **14** can be motionless as the outer discs **32**, **34** rotate in opposite directions. FIG. 5 illustrates the inner disc **14** of the wheel **10** rotating relative to both of the outer discs **32**, **34**. As shown in FIG. 5, the inner disc **14** can be rotated as the outer discs **32**, **34** remain motionless.

(17) Regardless of the absolute motion of the inner disc **14** and the outer discs **32**, **34**, the relative motion of the discs **14**, **32**, **34** changes the wheel **10** from a first configuration, as shown in FIGS. 6-7, to a second configuration, as shown in FIG. 8-9. In the first configuration, the curves of the serpentine isolator ring **48** are disposed in a first position relative to the second plurality of magnets **30** to isolate the first plurality of magnets **28** from the second plurality of magnets **30**, thereby generating a first magnetic flux **50** between the second plurality of magnets and the ferromagnetic surface **12** to increase the adhesion of the wheel to the ferromagnetic surface **12**. A weaker magnetic flux **52** is also generated. An air gap **54** is formed between the inner disc **14** and the ferromagnetic surface **12**.

(18) In the second configuration, at least one outer annular disc **32**, **34** is rotated about the axle relative to the inner annular disc **14** to dispose the curves of the serpentine isolator ring **48** in a second position relative to the second plurality of magnets **30** to allow magnetic interaction between the first plurality of magnets **28** and the second plurality of magnets **30**, thereby generating a second magnetic flux **56** between the second plurality of magnets **30** and the ferromagnetic surface **12** to decrease the adhesion of the wheel **10** to the ferromagnetic surface **12**. A stronger magnetic flux **58** is also generated between the magnets **28**, **30**. However, the second magnetic flux **56** is less than the first magnetic flux **50** shown in FIG. 7, so there is less adhesion of the wheel **10** to the ferromagnetic surface **12**.

(19) As shown in FIG. 10, a method **100** includes the step of providing the wheel **10** in step **110**, with the wheel **10** having a first annular disc retaining first and second magnets, and having a second annular disc with a serpentine isolator ring. The method **100** then disposes the first and second annular discs in a first configuration in step **120** with curves of the serpentine isolator ring in a first position relative to the magnets. The method **100** then isolates the first magnets from the second magnets in step **130**, and generates a first magnetic flux between the second magnets and a ferromagnetic surface in step **140** to increase the adhesion of the wheel to the ferromagnetic surface. The method **100** then rotates the second annular disc relative to the first annular disc to be in a second configuration in step **150**, and disposes the curves of the serpentine isolator ring in a second position in step **160**. The method **100** then allows magnetic interaction between the first and second magnets in step **170**, and generates a second magnetic flux between the second magnets and the ferromagnetic surface in step **180**, with the second magnetic flux being less than the first magnetic flux. The method **100** then decreases the adhesion of the wheel to the ferromagnetic surface in step **190**.

(20) Portions of the methods described herein can be performed by software or firmware in machine readable form on a tangible (e.g., non-transitory) storage medium. For example, the software or firmware can be in the form of a computer program including computer program code adapted to cause the system to perform various actions described herein when the program is run

on a computer or suitable hardware device, and where the computer program can be embodied on a computer readable medium. Examples of tangible storage media include computer storage devices having computer-readable media such as disks, thumb drives, flash memory, and the like, and do not include propagated signals. Propagated signals can be present in a tangible storage media. The software can be suitable for execution on a parallel processor or a serial processor such that various actions described herein can be carried out in any suitable order, or simultaneously.

(21) It is to be further understood that like or similar numerals in the drawings represent like or similar elements through the several figures, and that not all components or steps described and illustrated with reference to the figures are required for all embodiments or arrangements.

(22) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the invention. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “contains,” “containing,” “includes,” “including,” “comprises,” and/or “comprising,” and variations thereof, when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

(23) Terms of orientation are used herein merely for purposes of convention and referencing and are not to be construed as limiting. However, it is recognized these terms could be used with reference to an operator or user. Accordingly, no limitations are implied or to be inferred. In addition, the use of ordinal numbers (e.g., first, second, third) is for distinction and not counting. For example, the use of “third” does not imply there is a corresponding “first” or “second.” Also, the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting. The use of “including,” “comprising,” “having,” “containing,” “involving,” and variations thereof herein, is meant to encompass the items listed thereafter and equivalents thereof as well as additional items.

(24) While the disclosure has described several exemplary embodiments, it will be understood by those skilled in the art that various changes can be made, and equivalents can be substituted for elements thereof, without departing from the spirit and scope of the invention. In addition, many modifications will be appreciated by those skilled in the art to adapt a particular instrument, situation, or material to embodiments of the disclosure without departing from the essential scope thereof. Therefore, it is intended that the invention not be limited to the particular embodiments disclosed, or to the best mode contemplated for carrying out this invention, but that the invention will include all embodiments falling within the scope of the appended claims.

(25) The subject matter described above is provided by way of illustration only and should not be construed as limiting. Various modifications and changes can be made to the subject matter described herein without following the example embodiments and applications illustrated and described, and without departing from the true spirit and scope of the invention encompassed by the present disclosure, which is defined by the set of recitations in the following claims and by structures and functions or steps which are equivalent to these recitations.

Claims

1. A wheel having an axle and configured to adhere magnetically to a ferromagnetic surface, comprising: an inner annular disc composed of a non-magnetic material and having: a first outer circumferential periphery; a first central axial bore configured to retain the axle; a first plurality of apertures disposed adjacent to the first central axial bore and configured to retain a first plurality of magnets; and a second plurality of apertures disposed adjacent to the outer circumferential periphery and configured to retain a second plurality of magnets; and a pair of outer annular discs composed of a ferromagnetic material and disposed on either side of the inner annular disc, with

each outer annular disc having: a second outer circumferential periphery; a second central axial bore configured to retain the axle; an inner circumferential periphery disposed adjacent to the second central axial bore; and an isolator ring composed of a non-magnetic material and defining a plurality of curves extending in a serpentine manner circumferentially around the inner circumferential periphery intermediate of the second outer circumferential periphery and the second central axial bore, with the serpentine curves of the isolator ring disposed between the inner circumferential periphery and the second outer circumferential periphery, wherein in a first configuration, the serpentine curves of the isolator ring are disposed in a first position relative to the second plurality of magnets to isolate the first plurality of magnets from the second plurality of magnets, thereby generating a first magnetic flux between the second plurality of magnets and the ferromagnetic surface to increase the adhesion of the wheel to the ferromagnetic surface, and wherein in a second configuration, at least one outer annular disc is rotated about the axle relative to the inner annular disc to dispose the serpentine curves of the isolator ring in a second position relative to the second plurality of magnets to allow magnetic interaction between the first plurality of magnets and the second plurality of magnets, thereby generating a second magnetic flux between the second plurality of magnets and the ferromagnetic surface to decrease the adhesion of the wheel to the ferromagnetic surface, wherein the second magnetic flux is less than the first magnetic flux.

2. The wheel of claim 1, wherein each of the first plurality of magnets has a first polarity, and wherein each of the second plurality of magnets has a second polarity opposite to the first polarity.
3. The wheel of claim 1, wherein in the second configuration, the first plurality of magnets at least partially cancels the magnetic flux of the second plurality of magnets.
4. The wheel of claim 1, wherein the first plurality of magnets are permanent magnets.
5. The wheel of claim 1, wherein the second plurality of magnets are permanent magnets.
6. The wheel of claim 1, wherein the first and second pluralities of apertures are cylindrical.
7. The wheel of claim 6, wherein the first and second pluralities of magnets are cylindrical.
8. A wheel having an axle and configured to adhere magnetically to a ferromagnetic surface, comprising: a first annular disc composed of a non-magnetic material and having: a first outer circumferential periphery; a first central axial bore configured to retain the axle; a first plurality of apertures disposed adjacent to the first central axial bore and configured to retain a first plurality of magnets; and a second plurality of apertures disposed adjacent to the outer circumferential periphery and configured to retain a second plurality of magnets; and a second annular disc composed of a ferromagnetic material and disposed on one side of the first annular disc, with the outer annular disc having: a second outer circumferential periphery; a second central axial bore configured to retain the axle; an inner circumferential periphery disposed adjacent to the second central axial bore; and an isolator ring composed of a non-magnetic material and defining a plurality of curves extending in a serpentine manner circumferentially around the inner circumferential periphery intermediate of the second outer circumferential periphery and the second central axial bore, with the serpentine curves of the isolator ring disposed between the inner circumferential periphery and the second outer circumferential periphery, wherein in a first configuration, the serpentine curves of the isolator ring are disposed in a first position relative to the second plurality of magnets to isolate the first plurality of magnets from the second plurality of magnets, thereby generating a first magnetic flux between the second plurality of magnets and the ferromagnetic surface to increase the adhesion of the wheel to the ferromagnetic surface, and wherein in a second configuration, the second annular disc is rotated about the axle relative to the first annular disc to dispose the serpentine curves of the isolator ring in a second position relative to the second plurality of magnets to allow magnetic interaction between the first plurality of magnets and the second plurality of magnets, thereby generating a second magnetic flux between the second plurality of magnets and the ferromagnetic surface to decrease the adhesion of the wheel to the ferromagnetic surface, wherein the second magnetic flux is less than the first magnetic flux.

9. The wheel of claim 8, wherein each of the first plurality of magnets has a first polarity, and wherein each of the second plurality of magnets has a second polarity opposite to the first polarity.
 10. The wheel of claim 8, wherein in the second configuration, the first plurality of magnets at least partially cancels the magnetic flux of the second plurality of magnets.
 11. The wheel of claim 8, wherein the first plurality of magnets are permanent magnets.
 12. The wheel of claim 8, wherein the second plurality of magnets are permanent magnets.
 13. The wheel of claim 8, wherein the first and second pluralities of apertures are cylindrical.
 14. The wheel of claim 13, wherein the first and second pluralities of magnets are cylindrical.
 15. A method of adhering a wheel magnetically to a ferromagnetic surface, comprising: providing a wheel having an axle, a first annular disc, and a second annular disc, wherein the first annular disc is composed of a non-magnetic material and having: a first outer circumferential periphery; a first central axial bore configured to retain the axle; a first plurality of apertures disposed adjacent to the first central axial bore and configured to retain a first plurality of magnets; and a second plurality of apertures disposed adjacent to the outer circumferential periphery and configured to retain a second plurality of magnets; and wherein the second annular disc is composed of a ferromagnetic material and disposed on one side of the first annular disc, with the outer annular disc having: a second outer circumferential periphery; a second central axial bore configured to retain the axle; an inner circumferential periphery disposed adjacent to the second central axial bore; and an isolator ring composed of a non-magnetic material and defining a plurality of curves extending in a serpentine manner circumferentially around the inner circumferential periphery intermediate of the second outer circumferential periphery and the second central axial bore, with the serpentine curves of the isolator ring disposed between the inner circumferential periphery and the second outer circumferential periphery; disposing the first and second annular discs in a first configuration wherein the serpentine curves of the isolator ring are disposed in a first position relative to the second plurality of magnets; isolating the first plurality of magnets magnetically from the second plurality of magnets; and generating a first magnetic flux between the second plurality of magnets and the ferromagnetic surface thereby increasing the adhesion of the wheel to the ferromagnetic surface.
 16. The method of claim 15, further comprising: disposing the first and second annular discs in a second configuration, wherein the second annular disc is rotated about the axle relative to the first annular disc; disposing the serpentine curves of the isolator ring in a second position relative to the second plurality of magnets; allowing magnetic interaction between the first plurality of magnets and the second plurality of magnets; and generating a second magnetic flux between the second plurality of magnets and the ferromagnetic surface to decrease the adhesion of the wheel to the ferromagnetic surface, wherein the second magnetic flux is less than the first magnetic flux.
 17. The method of claim 15, wherein each of the first plurality of magnets has a first polarity, and wherein each of the second plurality of magnets has a second polarity opposite to the first polarity.
 18. The method of claim 15, wherein the first plurality of magnets are permanent magnets.
 19. The method of claim 15, wherein the second plurality of magnets are permanent magnets.
 20. The method of claim 15, wherein the first and second pluralities of apertures are cylindrical.
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